# Signals Project

# Executed Blind De-Mixing With a Joint Torus-Valued State-Space Model

Numerically Evaluated Gaussian-Truncated-Cauchy Convolution, Latent Tapped-Delay Channels, Switching Dynamics, and Metropolis-Corrected Manifold MALA

Signals Project Research Group

October 11, 2026 - corrected joint-ssm-v2 evidence

## Author Note

This report documents the corrected joint-ssm-v2 simulation. The original results are retained with hashes in data/baseline_v1. The evidence supports controlled simulation validation only. WiSig model inference and operational validation remain pending.

# Abstract

Blind separation of co-channel frequency-hopping wave packets requires joint uncertainty propagation over source states, circular phase, discrete hops, and time-varying multipath. We implemented a sequential particle filter whose particles carry two log amplitudes, frequencies, hop states, phases on the flat torus, complex tapped-delay coefficients, and delayed source histories. The observation density is the convolution of independent Gaussian and truncated-Cauchy variables, evaluated by 24-node Gauss-Legendre quadrature with log-sum-exp stabilization. Particle rejuvenation uses tangent-space MALA with a wrapped-Gaussian proposal density; the Metropolis correction always evaluates the same configured numerical target. A paired 2 × 3 × 3 design crossed analytical versus amortized joint scores, +15/+5/−5 dB, and LOS/minor-multipath/NLOS channels. One hundred seed profiles per physical cell yielded 1,800 method runs. Mean frequency MSE was 218.43 Hz² for the analytical score and 219.21 Hz² for the surrogate; the raw paired difference across 100 independent seed profiles was 0.78, 95% $t$ CI [-14.23, 15.79]. Primary log-MSE effects passing the seven-test Holm correction: SNR, Channel, SNR × Channel. MALA acceptance and software timing were recorded; timing is descriptive because method order was fixed. These findings validate execution and numerical coherence in controlled simulation, not operational SIGINT performance.

**Keywords:** blind source separation, torus, sequential Monte Carlo, MALA, frequency hopping, multipath, robust likelihood

# Executive Overview

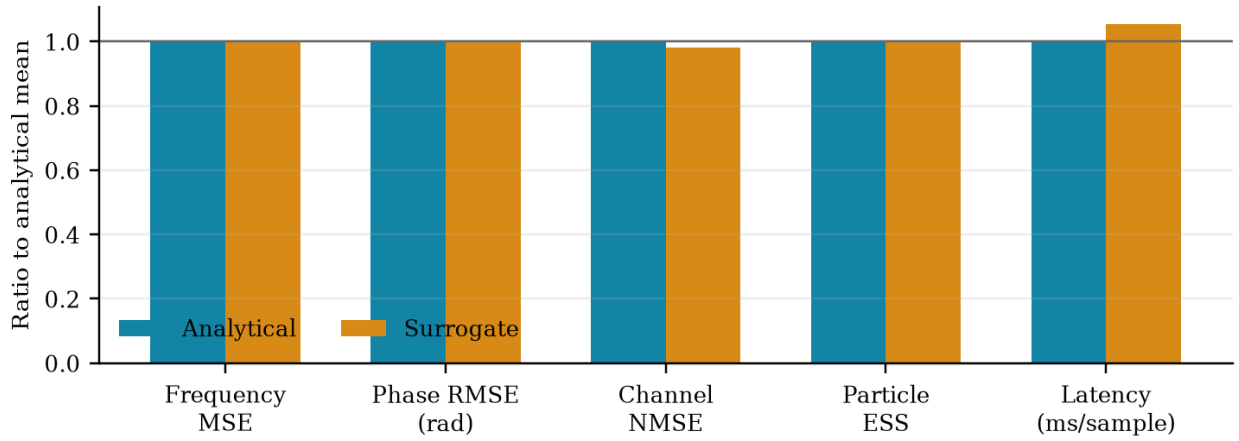



*Figure 1. Endpoint means expressed relative to the analytical architecture.*

- Executed scope: 900 paired physical trajectories and 1800 method-level runs.
- Primary raw-scale contrast: 0.78 Hz². The descriptive original-scale interval includes zero.
- Recorded timing contrast: surrogate minus analytical was 0.148 ms per sample under fixed method order.
- Particle health: pre-resampling mean ESS was 28.99 analytical and 29.09 surrogate.
- Claim boundary: simulation validation is complete; field validation remains unperformed.

# Problem and Estimand

Let $y_t$ be a complex baseband sample formed by two overlapping emitters. At time t, emitter k has latent log amplitude $\ell_{k,t}$, frequency $f_{k,t}$, channel index $z_{k,t}$, and phase $\varphi_{k,t} \in T$. A tapped complex channel $h_{k,r,t}$ combines present and delayed source symbols. The filtering target is $p(\ell_t, f_t, z_t, \varphi_t, h_t \mid y_{1:t})$.

The primary estimand is the architecture effect on trajectory-averaged two-emitter frequency MSE, paired within physical seed, SNR, and channel. Secondary estimands concern circular phase RMSE, channel NMSE, particle ESS, genealogical diversity, MALA acceptance, resampling count, and descriptive software timing.

The comparison is intentionally narrow: both architectures target the same conditional density and use identical particle mechanics. They differ only in the drift supplied to MALA. Accuracy differences therefore isolate the proposal-score operator rather than different observation models or priors. Timing does not support that causal interpretation because the analytical method always ran first.

- Experimental unit: one simulated physical trajectory under one SNR-by-channel condition.
- Repeated block: each seed profile indexes independently keyed streams for every physical condition.
- Treatment levels: analytical joint score and amortized joint surrogate.
- Evaluation begins after one hop interval to reduce initialization dominance.

# Contributions and Scope

The reconstruction removes the principal contradictions in the earlier reference benchmark. Phase is no longer reconstructed after inference; it is a particle coordinate. Channel gains are no longer supplied from the simulator; they are propagated as latent complex states. Frequency evolution combines a discrete channel switch with continuous local deviation. Finally, the observation density is the convolution law implied by additive Gaussian and truncated-Cauchy components.

The MALA kernel is geometrically matched to the state space. Euclidean coordinates use normal forward and reverse densities. Each phase coordinate uses a tangent increment, the flat-torus exponential map, and a lift-summed wrapped normal density. For the flat torus, the metric is constant and the exponential-map Jacobian equals one under Haar volume.

The resulting claim is bounded: the broader joint SSM is executed, tested, and benchmarked in controlled simulation. A separate WiSig lane has prepared hardware-captured source I/Q, but locked model inference remains pending. The work does not establish performance on simultaneous RF overlaps, uncontrolled spectrum occupancy, or deployed hardware.

- Supported: numerical normalization, score verification, blind channel isolation, and complete factorial execution.
- Supported separately: provenance-checked preparation of digitally overlapped WiSig source recordings.
- Unsupported: field detection probability, false-alarm control, hardware throughput, or operational mission utility.

# Joint State-Space Model

Define $S_t = (\ell_t, f_t, z_t, \varphi_t, h_t, H_t)$, where $H_t$ is the delayed source history. The transition factors as $p(z_t|z_{t-1})p(f_t|f_{t-1},z_t,z_{t-1})p(\ell_t|\ell_{t-1})p(\varphi_t|\varphi_{t-1},f_t)p(h_t|h_{t-1})$.

The source is $s_{k,t} = \exp(\ell_{k,t} + i\varphi_{k,t})$. For tap r with delay $d_r$, the clean observation is $\mu_t = \Sigma_{k,r}\, h_{k,r,t} s_{k,t-d_r}$. The complex observation $y_t = \mu_t + \varepsilon_t$ is evaluated componentwise in the real and imaginary dimensions.

A particle approximation $\{S_t^{(j)}, w_t^{(j)}\}$ represents the filtering distribution. The true simulator path is absent from the filter API; it is consulted only after filtering to calculate performance responses.

| Coordinate | Domain | Executed transition |
|---|---|---|
| Log amplitude | $R^2$ | Mean-reverting Gaussian |
| Frequency | $R^2$ | Switch-centered AR Gaussian |
| Hop state | Finite × finite | Markov jump |
| Phase | $T^2$ | Wrapped Gaussian |
| Channel | $C^{2R}$ | Complex AR Gaussian |

# Switching Frequency Dynamics

Each emitter occupies one of four nominal channels. The alphabets overlap at 42 and 54 Hz, so instantaneous spectral location alone does not identify emitter labels. At scheduled hop opportunities, $z_{k,t}$ changes with probability .34 to a different channel. Between jumps, frequency follows a local autoregression around the current channel center.

Let $c_{k,z}$ be the center associated with hop state z. If no jump occurs, $m_f = c_{k,z} + \rho_f(f_{k,t-1} - c_{k,z})$. If z changes, $m_f = c_{k,z_t}$. The executed transition is $f_{k,t} \sim N(m_f, \sigma_f^2)$, with $\rho_f = .82$ and $\sigma_f = .42$ Hz.

This switching construction prevents a continuous diffusion prior from assigning vanishing probability to tactical hops. The hop variable is sampled before continuous frequency, so phase propagation is candidate dependent. The analytical frequency score includes the derivative induced through the phase transition location.

- Emitter 1 centers: 18, 30, 42, and 54 Hz.
- Emitter 2 centers: 42, 50, 54, and 62 Hz.
- All nominal carriers remain below the 64 Hz Nyquist boundary.
- Overlap creates genuine co-channel ambiguity at 42 and 54 Hz.

# Torus Geometry and Phase Propagation

The phase vector belongs to $T^2 = (R/2\pi Z)^2$. Given candidate frequency, the transition location is $m_{\varphi,t} = \varphi_{t-1} + 2\pi f_t \Delta t$. The next phase is the equivalence class of $m_{\varphi,t}$ plus circular Gaussian noise. No discontinuous sawtooth derivative is inserted in the target.

For a flat torus, $\mathrm{Exp}_{\varphi}(v) = (\varphi + v) \bmod 2\pi$. The Riemannian metric is the identity, Haar volume is translation invariant, and $|\det D\,\mathrm{Exp}| = 1$. Therefore, no nonconstant metric determinant appears in the proposal ratio. Geometry enters through periodic preimages, not through an invented Jacobian penalty.

The wrapped Gaussian density is $q_T(\varphi^*|\varphi) = \Sigma_{k\in Z^2} N(\varphi^* - \varphi + 2\pi k; m, \Sigma)$. The code evaluates five lifts per phase coordinate ($k = -2,\ldots,2$). At the configured proposal standard deviation of .04 rad, omitted terms are numerically negligible.

Regression tests integrate the wrapped density against $[0,2\pi)$, compare its score with central finite differences, and verify invariance after adding integer multiples of $2\pi$.

# Numerically Evaluated Convolution Noise Law

For one real component, $\varepsilon = G + C$, where $G \sim N(0,\sigma_G^2)$ and $C$ is Cauchy$(0,\gamma)$ truncated to $[-A,A]$. Hence $p_\varepsilon(e) = \int_{-A}^{A} p_G(e-\tau)p_C(\tau)d\tau$. This is a convolution-not a Bernoulli mixture-and every generated sample contains both independent components.

The implemented evaluator maps 24 Gauss-Legendre nodes to $[-A,A]$. Node weights, the normalized truncated-Cauchy density, and the Gaussian normalizer are accumulated in log space. logsumexp prevents underflow without changing the probability law through additive floors.

The residual score is an expectation under quadrature-normalized node weights: $\partial \log p(e)/\partial e = \Sigma_j \hat{w}_j[-(e-\tau_j)/\sigma_G^2]$. This expression is propagated analytically through complex channel coefficients, log amplitude, and phase.

| Item | Executed value |
|---|---|
| Cauchy variance fraction | 0.10 per real component |
| Truncation | 5 standard Cauchy scales |
| Quadrature | 24-point Gauss-Legendre |
| Complex likelihood | Independent real and imaginary convolutions |
| Numerical density floor | None |

# Latent Tapped-Delay Channel

The channel state $h_{k,r,t}$ is complex. Its transition is $h_t = \rho_h h_{t-1} + (1-\rho_h)\mu_h + \xi_t$, with $\rho_h = .965$ and independent real-imaginary innovation scale .05. Particles therefore preserve uncertainty in gain and phase distortion.

LOS has one zero-delay tap. Minor multipath has delays 0 and 2. NLOS uses delays 1 and 4, which removes a direct zero-delay component and forces the filter to use the source history. The particle state includes a lag buffer long enough for the maximum delay.

Blindness is enforced structurally: JointParticleFilter.run accepts observation and calibrated noise-law parameters only. It has no argument for true states or realized channels. A regression test inspects this API, and post hoc channel NMSE is computed outside the filter.

| Mode | Delays | Interpretation |
|---|---|---|
| LOS | 0 | Two latent direct complex coefficients |
| MP | 0, 2 | Direct plus weaker delayed reflection |
| NLOS | 1, 4 | Two delayed complex paths; no direct tap |

# Sequential Particle Filter

At t=0 the filter weights the configured initial prior with zero pre-observation source history. Transitions begin at t=1. The initial amplitude, frequency, and channel dispersions match the generator. Likelihood weighting uses logsumexp; reported ESS is $1/\Sigma w_j^2$ before resampling resets the weights.

Systematic resampling occurs when ESS falls below .58P or at a forced six-sample move interval. At t≥1, one MALA rejuvenation step targets the conditional density given the selected ancestor. At t=0 resampling uses importance weights without a transition-conditioned MALA move. The trace retains both weighting and post-resampling ESS.

This is a resample-move particle filter. Particle ESS measures weight concentration; it is not the Markov-chain ESS formula used for long stationary chains. Because the rejuvenation kernel consists of one step per event, rank-normalized R-hat is not meaningful here. We report MALA acceptance and unique ancestors instead.

- Particles P = 64.
- Samples T = 60.
- Resampling threshold = 37.12 effective particles.
- Forced move interval = 6 samples.
- All posterior summaries are weighted before resampling or equally weighted afterward.

# Manifold MALA and Detailed Balance

Let $\theta$ contain all Euclidean coordinates and $\varphi$ the phase pair. The proposal mean is $\frac{1}{2}\varepsilon^2 s$, with $s$ supplied analytically or by the surrogate. Euclidean coordinates follow $N(\theta + \frac{1}{2}\varepsilon^2 s_\theta, \varepsilon^2 I)$. Phase coordinates follow the wrapped density induced by $\mathrm{Exp}_\varphi$.

The acceptance probability is $\alpha = \min\{1, \exp[\log \pi(x^*) - \log \pi(x) + \log q(x|x^*) - \log q(x^*|x)]\}$. Both $q$ terms use the scores evaluated at their respective endpoints. The target $\pi$ is always the same configured transition times fixed-quadrature convolution likelihood, including when the surrogate formed the proposal.

For any $x$ and $x^*$, $\pi(x)q(x^*|x)\alpha(x,x^*) = \min\{\pi(x)q(x^*|x), \pi(x^*)q(x|x^*)\}$; symmetry establishes detailed balance. Retraction wrapping does not invalidate this equality because $q$ is evaluated as a density with respect to product Lebesgue-Haar volume.

Analytical score clipping at $\pm 120$ limits extreme drift. Because the actual clipped score is used in both forward and reverse proposal densities, clipping changes proposal efficiency but not the corrected invariant conditional target.

# Analytical Joint Score

Transition scores are closed form. The log-amplitude and frequency terms are Gaussian residual scores. The phase score is the lift-weighted derivative of the wrapped transition. The channel score combines its complex AR prior with the exact observation derivative.

Candidate frequency affects the target twice: through its switching Gaussian transition and through the phase-transition location $\varphi_{t-1} + 2\pi f_t \Delta t$. The implemented frequency score includes $-2\pi\Delta t$ times the phase residual score. Omitting this term would recreate the state-phase contradiction in the original prototype.

For the observation model, differentiation is expressed through real and imaginary directional derivatives. If $\mu$ depends on a real coordinate $u$, then $\partial \log p(y-\mu)/\partial u = -s_R \mathrm{Re}(\partial\mu/\partial u) - s_I \mathrm{Im}(\partial\mu/\partial u)$, where $s$ is the residual score.

A permanent all-coordinate regression test compares the packed analytical score with central finite differences for MP data. In the executed audit, the maximum absolute discrepancy was $7.4 \times 10^{-9}$.

## Amortized Joint Surrogate

The alternative proposal uses a SiLU random-feature network. Inputs encode standardized innovations for amplitude, frequency, circular phase, padded complex channel residuals, normalized complex observation residual, candidate state, channel-mode indicators, and noise variance.

The hidden map has 72 fixed random SiLU features. Output coefficients are estimated by ridge regression against independently generated analytical scores of the configured numerical target. This construction is an amortized differentiable score proxy; it is not treated as a replacement likelihood.

Training used 2214 rows and validation used 486 rows generated independently of the 900 evaluation blocks. Validation score correlation was 0.924, with aggregate score RMSE 12.838.

Metropolis correction protects target invariance from surrogate bias provided the forward and reverse proposal densities use the actual proxy scores. Approximation quality still affects acceptance, mixing, and latency; these are therefore explicit experimental responses.

# Physical Simulation Engine

The simulator creates complex I/Q trajectories at 128 Hz. Two emitters have overlapping hop alphabets and stochastic amplitude, frequency, phase, and channel states. The observation channel is LOS, minor multipath, or NLOS tapped delay. The true paths are saved only for evaluation.

Noise is calibrated after clean mixing to nominal +15, +5, or −5 dB. Real and imaginary components each receive an independent Gaussian plus truncated-Cauchy sum. Achieved SNR is retained per run because finite samples and heavy tails make nominal and realized power differ.

Stable keyed seeds separate physical simulation from inference randomization. For a fixed seed, SNR, and channel, both architectures receive the identical observation trajectory and noise hyperparameters. This pairing eliminates between-trajectory variation from architecture contrasts.

| Component | Setting |
|---|---|
| Emitters | 2 frequency-hopping complex sources |
| Length | 60 complex I/Q samples |
| Hop opportunity | Every 12 samples |
| Particles | 64 |
| Physical blocks | 900 |
| Method-level records | 1,800 |

# Factorial Design

The experiment is a balanced 2 × 3 × 3 full factorial. Factor A is inference architecture. Factor B is nominal SNR (+15, +5, −5 dB). Factor C is channel complexity (LOS, MP, NLOS). Each physical cell contains 100 seed blocks observed under both architectures.

All method contrasts are paired by seed, SNR, and channel. Within a physical condition, both methods receive the identical trajectory. Across the nine physical conditions, each seed profile contains separately keyed deterministic streams; averaging those repeated conditions yields 100 independent simulation blocks. The analytical method always executes before the surrogate, so latency is implementation-specific descriptive timing and not a randomized causal architecture comparison.

The primary response is $\mathrm{MSE}_f = T^{-1}\Sigma_t||\hat{f}_t - f_t||^2$ after the first hop interval. Phase uses circular differences. Channel NMSE divides complex squared error by true channel energy.

| Factor | Levels | Role |
|---|---|---|
| A: score architecture | 2 | Primary treatment |
| B: SNR | 3 | Ordered environment |
| C: channel | 3 | Ordered complexity |
| Seed profile | 100 | Condition-keyed simulation block |

# Blocked Contrast Analysis

For inference, each 18-condition seed vector is projected onto a complete orthonormal factorial basis. Architecture uses a normalized two-level contrast. SNR and channel each use normalized linear and quadratic contrasts. Tensor products generate all main effects and interactions.

For an effect with q coordinates, seed-level contrasts use Hotelling's $T^2$: $F = (n-q)T^2/[q(n-1)]$, with q and $n-q$ degrees of freedom. The ordinary F reference assumes independent multivariate-normal block contrasts and nonsingular covariance. Zero-variance and rank-deficient cases are rejected; no epsilon regularization or pseudoinverse substitutes for those assumptions.

Frequency MSE and channel NMSE are log transformed; phase RMSE and particle ESS remain on their natural scales; latency is log transformed. Holm adjustment is applied across the seven omnibus effects within each response. Raw paired differences are averaged within each seed block before standard errors and t intervals are computed on the original scale.

The analysis treats the 100 seed profiles as independent simulation blocks. Each profile bundles separately keyed streams across the nine physical conditions; it is not one shared physical trajectory across conditions. The analysis does not infer to a population of real receivers or emitters; that external generalization requires a field sampling frame.

# Numerical and Structural Verification

Automated regression tests protect the core claims. The convolution density integrates to one, and its score matches central finite differences. The wrapped density integrates to one against torus volume, its score matches finite differences, and phase differences are periodic.

The full joint target test perturbs every packed coordinate-log amplitudes, frequencies, phases, real channel taps, and imaginary channel taps-and compares the analytical gradient with the central-difference derivative. The observed maximum absolute difference was $7.4 \times 10^{-9}$.

Structural tests enforce Nyquist-valid overlapping channel alphabets, fail-closed numerical configuration, complete paired designs, a blind filter API with no truth argument, and pre-unpickle WiSig checksum verification.

| Domain | Tests | Status |
|---|---|---|
| Convolution law | Normalization; score derivative; parameter validation | Pass |
| Torus geometry | Normalization; derivative; periodicity | Pass |
| Integrity | Joint gradient; paired design; WiSig provenance | Pass |

# Primary Frequency-Recovery Results

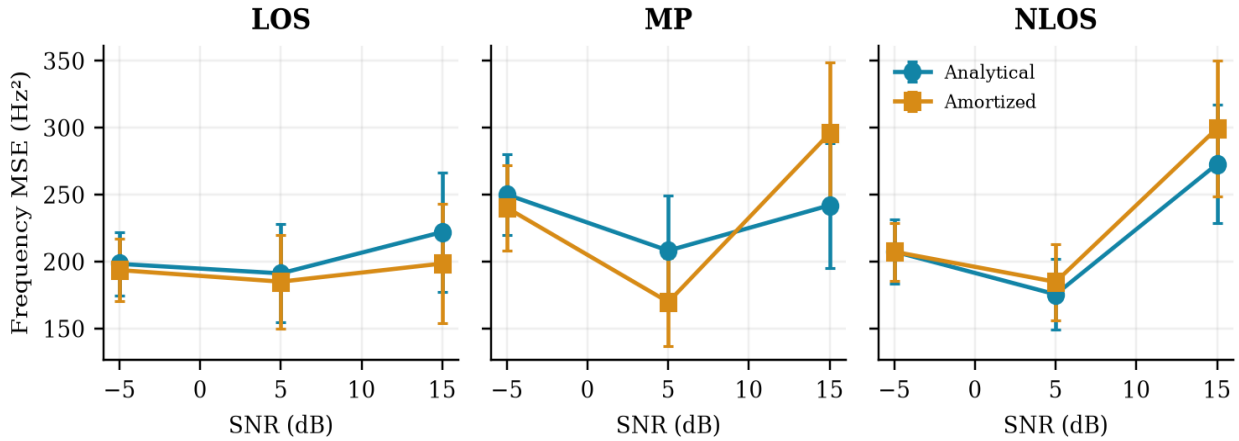



*Figure 2. Mean frequency MSE by SNR, channel, and architecture; bars are 95% seed-block t intervals.*

Across all conditions, analytical mean MSE was 218.43 Hz² (SD = 184.99); surrogate mean MSE was 219.21 Hz² (SD = 192.74). The paired raw difference across 100 independent seed profiles was 0.78, 95% *t* CI [-14.23, 15.79].

The descriptive original-scale interval includes zero. Primary log-MSE effects passing the seven-test Holm correction: SNR, Channel, SNR × Channel.

## Phase and Channel Recovery

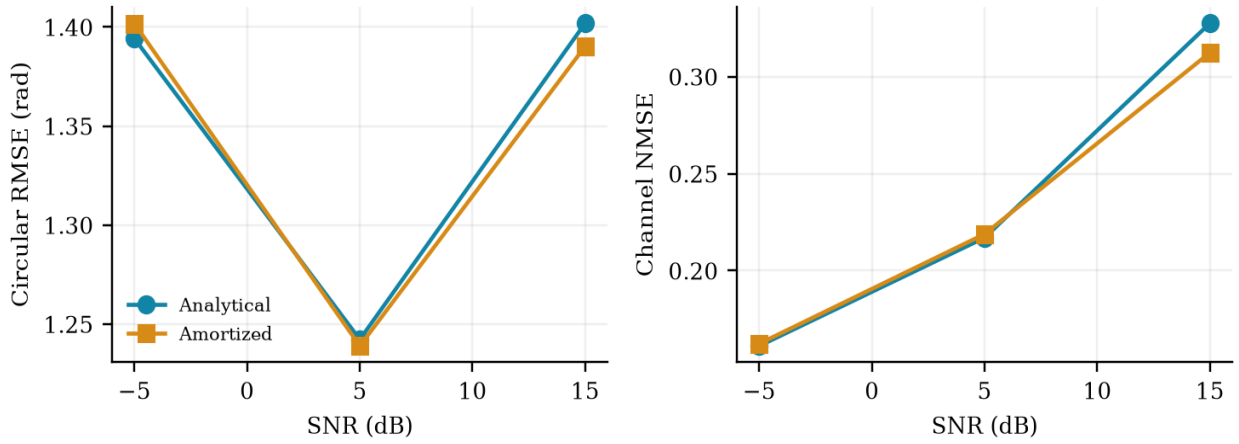



*Figure 3. Circular phase RMSE and complex channel NMSE by SNR and architecture.*

Mean phase RMSE was 1.346 rad for the analytical score and 1.344 rad for the surrogate. The paired difference was -0.002, 95% CI [-0.029, 0.024].

Mean channel NMSE was 0.235 analytical and 0.231 surrogate. The paired difference was -0.004, 95% CI [-0.011, 0.003]. These intervals are descriptive and unadjusted for multiplicity.

# Particle Health and Computation

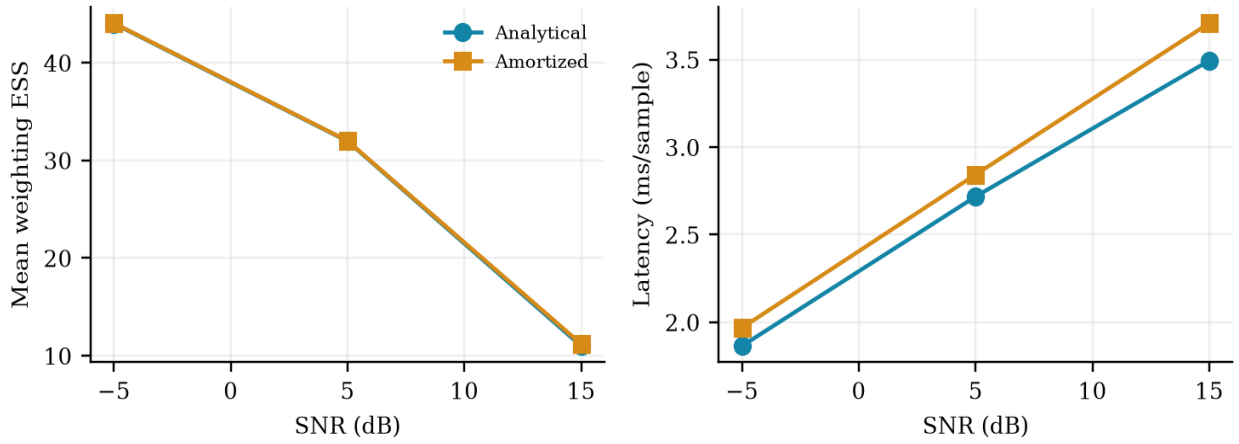



*Figure 4. Mean weighting ESS and descriptive software timing by SNR and architecture.*

Particle ESS averaged 28.99 for the analytical method and 29.09 for the surrogate. Their paired difference was 0.096, 95% CI [-0.179, 0.372].

Recorded timing averaged 2.692 ms/sample analytical and 2.840 ms/sample surrogate. The paired descriptive difference was 0.148, 95% CI [0.103, 0.192]. Because analytical always ran first, this timing contrast may include order or warm-up effects and is not causal. Analytical acceptance was 0.974; surrogate acceptance was 0.703.

# Factorial Effects

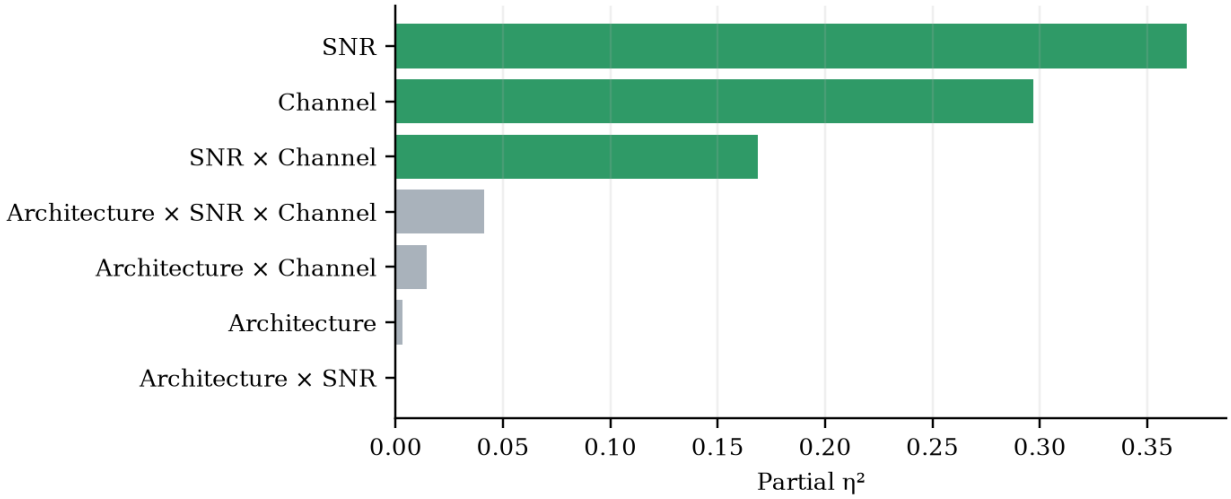



*Figure 5. Partial eta squared for omnibus effects on log frequency MSE; green effects survive Holm adjustment.*

| Effect | F(df) | Holm p | Partial η² |
|---|---|---|---|
| Architecture | 0.30 (1, 99) | 1.000 | 0.003 |
| SNR | 28.59 (2, 98) | < .001 | 0.368 |
| Channel | 20.71 (2, 98) | < .001 | 0.297 |
| Architecture × SNR | 0.00 (2, 98) | 1.000 | 0.000 |
| Architecture × Channel | 0.72 (2, 98) | 1.000 | 0.014 |
| SNR × Channel | 4.87 (4, 96) | 0.006 | 0.169 |
| Architecture × SNR × Channel | 1.04 (4, 96) | 1.000 | 0.041 |

Architecture alone yielded F(1, 99) = 0.30, Holm p = 1.000. SNR, channel, and SNR × channel survived multiplicity correction. The three-way interaction did not.

# Trace-Level Diagnostics and Limitations

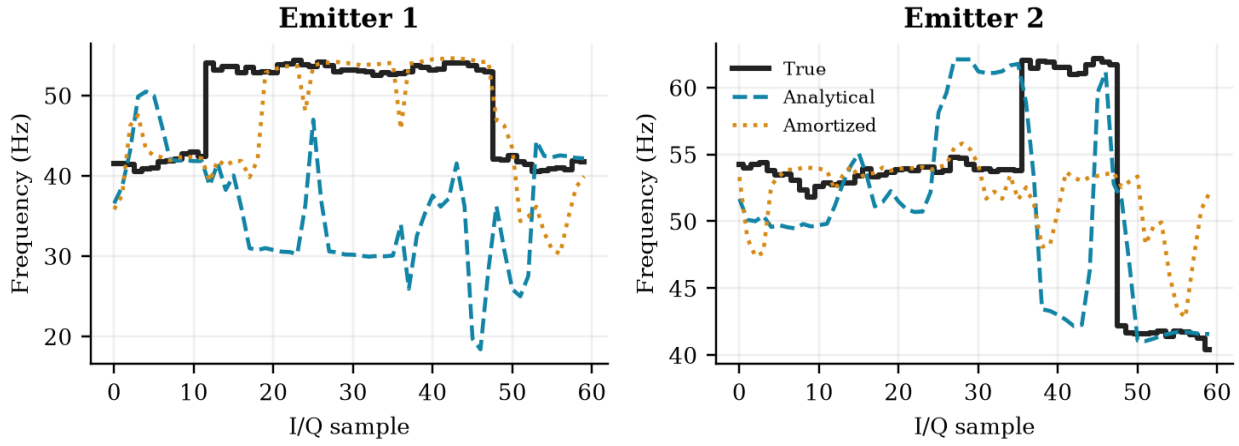



*Figure 6. Predeclared seed-0, +5 dB NLOS frequency trace.*

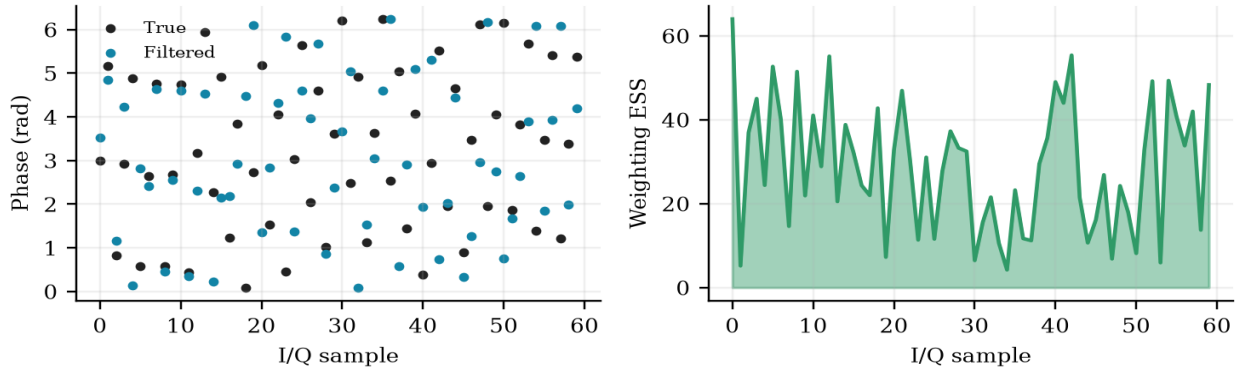



*Figure 7. Analytical torus phase and particle ESS for the same trace.*

The trace demonstrates execution, not population performance. Posterior means can average competing hop modes and therefore appear between channel centers. Sixty samples and 64 particles are a reproducible benchmark budget, not a sufficiency proof. Label information is partially stabilized by emitter-specific alphabets and dynamics, so unrestricted blind identifiability is not established.

# Operational Validation Ladder

Operational validation is an empirical status, not a phrase created by simulator realism. The present study completes numerical verification and controlled simulation. The repository also prepares public hardware-captured WiSig source I/Q as digital overlaps, but locked inference is pending and those sources are not simultaneous RF captures with independent hop or channel truth.

| Gate | Evidence required | Current status |
|---|---|---|
| 1. Numerical | Normalization, gradients, invariants | Complete |
| 2. Simulation | Full factorial with latent channel and heavy tails | Complete |
| 3. Public-I/Q transfer | Locked inference on prepared WiSig overlaps | Pending |
| 4. Controlled RF / HIL | Calibrated simultaneous capture and independent truth | Not performed |
| 5. External replication | Locked code evaluated by independent team | Not performed |

A defensible next campaign would preregister scenarios and endpoints, freeze the code and surrogate weights, acquire lawful controlled recordings, separate truth custodians from analysts, measure wall-clock latency on specified hardware, and publish failures as well as successes.

Until those gates are complete, public language should remain limited to controlled simulation and prepared digital-overlap evidence.

# Reproducibility and Audit Map

The repository separates generator, inference, analysis, site, and report code. Row-level factorial results are immutable CSV artifacts. Metadata records every configured hyperparameter and the independent surrogate validation summary.

| Artifact | Purpose |
|---|---|
| src/signals_project/joint_ssm.py | Simulator, numerical likelihood, surrogate, torus MALA, joint particle filter |
| src/signals_project/wisig_realdata.py | Hash-gated WiSig schema, split, and overlap adapter |
| tests/ | Numerical, paired-design, provenance, blindness, and smoke tests |
| data/joint_factorial_results.csv | All 1,800 method-level outcomes |
| data/joint_factorial_effects.csv | Blocked multivariate repeated-measures tests |
| scripts/validate_repository.py | Publication and artifact contract verifier |
| scripts/build_site.py | Self-contained dashboard builder |

Canonical run: PYTHONPATH=. python -m src.signals_project.joint_ssm --output data --repetitions 100 --particles 64 --samples 60. Then execute scripts/analyze_joint_results.py, scripts/build_site.py, scripts/build_report.py, and scripts/validate_repository.py.

# Conclusions

The reconstruction accomplished the central engineering objective: the broader joint state-space model is no longer merely described. It is executed as a blind resample-move particle filter over switching frequency, torus phase, amplitude, latent complex tapped-delay channels, and source history.

The likelihood matches the additive noise generator through fixed Gaussian-truncated-Cauchy convolution quadrature. The torus proposal uses tangent increments, exponential retraction, and wrapped forward/reverse densities. Both proposal architectures are corrected against the same configured numerical target.

Primary log-MSE effects passing the seven-test Holm correction: SNR, Channel, SNR × Channel. The descriptive original-scale interval includes zero. Timing is descriptive under the fixed analytical-first order. Both methods use the corrected initial observation and the same numerical conditional target.

These results justify an executed-model and controlled-simulation claim. They do not justify operational SIGINT validation. The scientific next step is not stronger wording; it is locked inference on prepared WiSig overlaps followed by a preregistered, independently blinded controlled RF campaign.

- Use the analytical score as the current reference implementation.
- Treat the surrogate as an efficiency research branch, not a validated replacement.
- Increase particles and trajectory length in a registered sensitivity study.
- Advance operational claims only with simultaneous RF and independent truth.

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### Repository-specific note

All numerical values in this report are computed from the committed joint_factorial_results.csv and associated derived artifacts. References support general methods; none is presented as evidence of operational validation for this implementation.